

Question

$\text{Fe}(\text{CO})_4$ is an unstable molecule, and crystals cannot be isolated for structure determination. Its molecular structure can only be studied through infrared spectroscopy, theoretical calculations, and other methods. Calculations show that the ground state $\text{Fe}(\text{CO})_4$ molecule has 4 unpaired electrons at normal pressure; under high pressure, it transforms into a diamagnetic molecule.

The following statements are made:

1. $\text{Fe}(\text{CO})_4$ satisfies the EAN rule
2. Under high pressure, the 4s orbital in the valence electron configuration of $\text{Fe}(\text{CO})_4$ is not filled with electrons
3. Under normal pressure, the coordination polyhedron of $\text{Fe}(\text{CO})_4$ is a regular tetrahedron
4. Applying pressure does not change the coordination polyhedron type of $\text{Fe}(\text{CO})_4$

The following option contains all correct statements and the most correct statements:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: C

Detailed Explanation

The number of electrons around iron is: $8 + 4 \times 2 = 16$, which does not satisfy the EAN rule, statement 1 is incorrect.

CHECKPOINT

1 PTS

The number of electrons around iron is 16, which does not conform to the EAN rule

Under high pressure, $\text{Fe}(\text{CO})_4$ transforms into a diamagnetic molecule, indicating that all electrons are paired. Therefore, the electronic configuration is $3d^84s^0$, and the $4s$ orbital is not filled with electrons, statement 2 is correct.

CHECKPOINT

1 PTS

Under high pressure, the electronic configuration of $\text{Fe}(\text{CO})_4$ is $3d^84s^0$

Under normal pressure, $\text{Fe}(\text{CO})_4$ has 4 unpaired electrons, corresponding to the high-spin d^6 configuration. The corresponding crystal field (or ligand field) splitting energy should be small, so it should be a tetrahedral type; the d^6 electronic configuration is asymmetrically filled in the electron orbitals of the regular tetrahedral field, and Jahn-Teller distortion will occur to transform into a distorted tetrahedron, statement 3 is incorrect.

CHECKPOINT

1 PTS

Under normal pressure, $\text{Fe}(\text{CO})_4$ corresponds to the high-spin d^6 configuration, which is asymmetrically filled in the electron orbitals of the regular tetrahedral field, and Jahn-Teller distortion will occur to transform into a distorted tetrahedron

Under high pressure, it is diamagnetic, corresponding to the low-spin d^8 configuration. The corresponding crystal field (or ligand field) splitting energy is large, so it should be a square planar type, statement 4 is incorrect.

CHECKPOINT

1 PTS

Under high pressure, the electronic configuration of $\text{Fe}(\text{CO})_4$ is $3d^8 4s^0$, the corresponding crystal field (or ligand field) splitting energy is large, so it should be a square planar type